

TF-IDF

Term Frequency – Inverse Document Frequency

Turning ingredient lists into comparable number arrays.

VECTORIZATION: TF-IDF

Problem: A computer can't compare "Chicken Parm" and "Eggplant Parm" – it can only compare numbers.

The idea: Turn each recipe into a list of numbers – one per ingredient. If an ingredient appears in the recipe, its number goes up.

Recipe A: ["Chicken", "Garlic", "Salt"]

Recipe B: ["Beef", "Garlic", "Salt"]

$$v_A = [\text{Chicken} : 1, \text{Garlic} : 1, \text{Salt} : 1]$$

VECTORIZATION: TF-IDF

Problem: A computer can't compare "Chicken Parm" and "Eggplant Parm" – it can only compare numbers.

The problem with raw counts:

"Salt" is in 99% of recipes. It drowns out everything else. Every recipe ends up pointing in roughly the same direction – toward "Salt."

$$v_A = \left[\text{Chicken} : 1, \text{Garlic} : 1, \underbrace{\text{Salt} : 1}_{\text{same in every recipe}} \right]$$

Result: recipes cluster by pantry staples, not by flavor.

TF-IDF: PENALIZING COMMON INGREDIENTS

Fix: scale down ingredients that appear everywhere. Scale up ingredients that are distinctive.

Inverse Document Frequency – the rarity penalty:

$$\text{IDF}(t) = \log \frac{N}{\text{DF}(t)}$$

N = total recipes. $\text{DF}(t)$ = how many recipes contain t .
Rarer ingredient = higher score.

- “Chicken” in 100/10,000 recipes $\rightarrow \text{IDF} = \log(100) = 2.0$
- “Salt” in 9,900/10,000 recipes $\rightarrow \text{IDF} = \log(1.01) \approx 0.004$

TF-IDF: PENALIZING COMMON INGREDIENTS

Fix: scale down ingredients that appear everywhere. Scale up ingredients that are distinctive.

Multiply TF × IDF:

$$v_A = [2.0, \\ 0.5, \\ 0.004]$$

```
>> execution_success
```

“Salt” is nearly zeroed out. “Chicken” dominates.

Two recipes with the same **distinctive** ingredients now point

in the same direction

– and K-NN can find them.

UMAP

Uniform Manifold Approximation and Projection
Compressing 5,000 dimensions into a renderable 3D space.

UMAP: THE DIMENSION PROBLEM

We have 5,000 ingredients. Each one is its own axis. That's 5,000 dimensions.

After TF-IDF, every recipe is a coordinate in $\mathbb{R}^{5000}$.
That's mathematically valid – but completely unrenderable.

$$M \in \mathbb{R}^{10\{,\}000 \times 5\{,\}000} \rightarrow \text{Three.js needs } \mathbb{R}^3$$

We need to compress 5,000 axes down to 3 **without scrambling which recipes are similar to which.**

UMAP: THE DIMENSION PROBLEM

We have 5,000 ingredients. Each one is its own axis. That's 5,000 dimensions.

Why not just drop 4,997 axes?

Any single axis (e.g. "Garlic") tells you almost nothing alone.

The **relationships between axes** is where the information lives.

Why not PCA?

PCA finds straight lines of variance. Flavor space is curved —

"Ramen" is close to both "Soups" and "Asian Noodles"

UMAP: MAPPING THE NEIGHBORHOOD STRUCTURE

UMAP records which recipes are neighbors in 5,000D, then rebuilds those relationships in 3D.

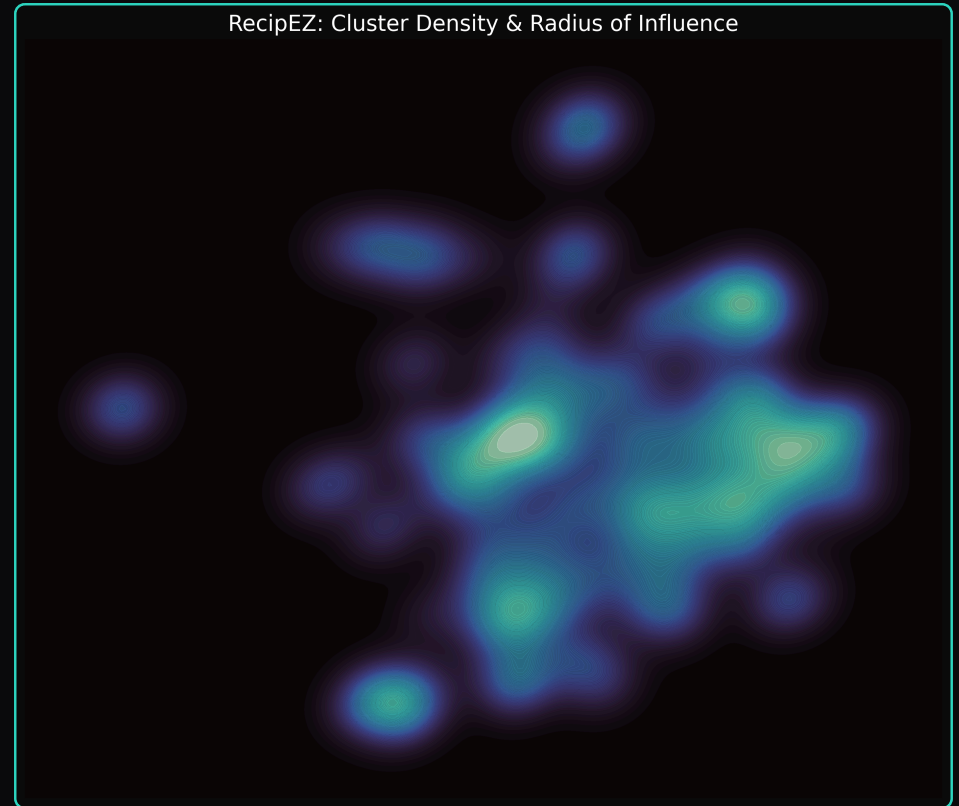
UMAP: MAPPING THE NEIGHBORHOOD STRUCTURE

UMAP records which recipes are neighbors in 5,000D, then rebuilds those relationships in 3D.

Step 1 – Local neighborhoods

For each recipe, find its k nearest neighbors ($k=15$). The distance to the closest one sets the local scale ρ_i :

$$\rho_i = \min_{j \in \text{kNN}(i)} d(i, j)$$



UMAP: MAPPING THE NEIGHBORHOOD STRUCTURE

UMAP records which recipes are neighbors in 5,000D, then rebuilds those relationships in 3D.

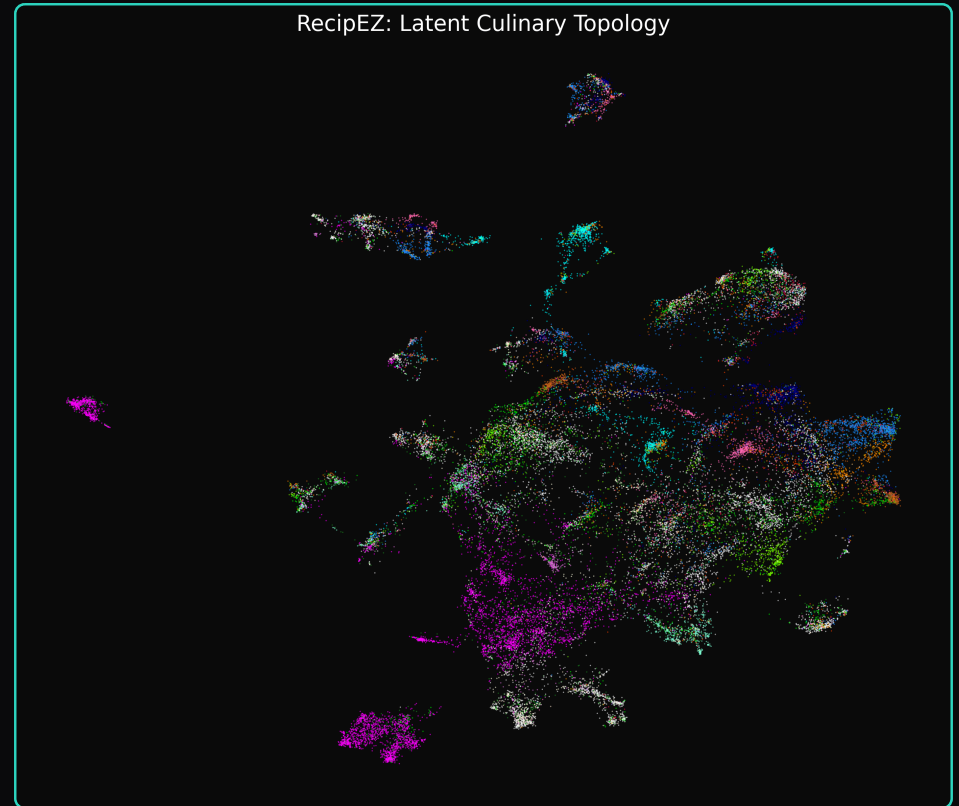
Step 2 – Fuzzy edges

Connections are weighted probabilities, not hard lines:

$$w_{ij} = \exp(-(d(i, j) - \rho_i)/\sigma_i)$$

Nearby recipes: $w \approx 1$

Distant recipes: $w \approx 0$



UMAP: REBUILDING THE MAP IN 3D

Recipes start at random 3D positions. Two forces push and pull them into the right shape.

Attraction – pulls neighbors together

$$F_{\text{att}} = \frac{-2abd^{2b-2}}{1 + ad^{2b}}$$

High- w pairs (similar recipes) get pulled close.

Repulsion – pushes non-neighbors apart

$$F_{\text{rep}} = \frac{2b}{(\epsilon + d^2)(1 + ad^{2b})}$$

Low- w pairs (unlike recipes) get pushed away.

200 optimization steps. Converges when the 3D map matches the 5,000D map.

Proximity in 3D space = culinary similarity.

UMAP: OUTPUT

Each recipe is now an (x, y, z) coordinate. Similar recipes are physically close.

$$v_{\text{ChickenParm}} = [+14.2, -3.8, +8.1]$$

$$v_{\text{EggplantParm}} = [+13.8, -3.5, +8.4]$$

$$v_{\text{ChocolateCake}} = [-22.1, +17.3, -5.9]$$

"Chicken Parm" ↔ "Eggplant Parm": **0.6 units apart**

"Chicken Parm" ↔ "Chocolate Cake": **37 units apart**

The geometry now reflects culinary reality.

UMAP: OUTPUT

Each recipe is now an (x,y,z) coordinate. Similar recipes are physically close.

```
>> render_ready
```

5,000 ingredient axes $\rightarrow$ 3 spatial coordinates.

Three.js can place every recipe as a point in a navigable 3D galaxy.

TF-IDF ✓ $\rightarrow$ UMAP ✓ $\rightarrow$ SVD $\rightarrow$ K-NN

SVD

Singular Value Decomposition

Discovering hidden flavor dimensions from ingredient co-occurrence.

SVD: FINDING HIDDEN FLAVOR GROUPS

TF-IDF treats every ingredient as independent. But ingredients travel in packs.

Basil, Oregano, Tomato, and Mozzarella almost always appear together.

They're not four separate axes of information – they're one concept: **Italian**.

TF-IDF has no way to know this. The 5,000D space is full of redundant, correlated dimensions.

Consequence: K-NN is comparing noise alongside signal.

SVD: FINDING HIDDEN FLAVOR GROUPS

TF-IDF treats every ingredient as independent. But ingredients travel in packs.

SVD discovers these “flavor axes” automatically by analyzing which ingredients co-occur across the whole corpus.

$$M_{10\{\},\}000 \times 5\{\},\}000 \rightarrow \text{SVD} \rightarrow \tilde{M}_{10\{\},\}000 \times 128$$

5,000 raw ingredient axes $\rightarrow$ 128 **latent flavor dimensions**.
91% of the information. 2.5% of the dimensions.

SVD: THE DECOMPOSITION

Any matrix can be factored into three simpler matrices.

$$M = U\Sigma V^T$$

U

Shape: $N \times N$

Each row = one
recipe as a mix of
flavor dimensions.

$$U^T U = I$$

Σ

Diagonal.

$$\sigma_1 \geq \sigma_2 \geq \dots \geq 0$$

Each value = how
much variance that
flavor dimension
explains.

Sorted by importance.

V^T

Shape: $D \times D$

Each row = the
ingredient blend
for one flavor
dimension.

$$V V^T = I$$

SVD: KEEPING ONLY WHAT MATTERS

Keep the top k dimensions. Drop the rest – they're noise.

$$M \approx \tilde{M} = U_k \Sigma_k V_k^T$$

$$\|M - \tilde{M}\|_F = \sqrt{\sum_{i=k+1}^r \sigma_i^2} \rightarrow \text{minimized}$$

The Eckart-Young theorem guarantees this is the **best possible** rank- k approximation.

At $k=128$: variance explained = **91.3%**, dimensions dropped = **97.4%**.

The small σ_i values we discard encode coincidences – not real culinary relationships.
We keep the signal. We discard the noise.

SVD: WHAT THE FLAVOR DIMENSIONS LOOK LIKE

The rows of V_k^T are not hand-labeled – the algorithm discovers these groupings on its own.

Dim 1 ($\sigma=142$): Basil +0.41 · Tomato +0.38 · Oregano +0.35 · Mozzarella +0.29
→ Italian / Mediterranean

Dim 2 ($\sigma=119$): Soy Sauce +0.44 · Ginger +0.39 · Sesame +0.31 · Rice Wine +0.28
→ East Asian

Dim 3 ($\sigma=97$): Cumin +0.42 · Chili +0.40 · Coriander +0.33 · Lime +0.27
→ South Asian / Mexican

Dim 4 ($\sigma=84$): Butter +0.45 · Sugar +0.43 · Vanilla +0.38 · Flour +0.35
→ Desserts / Baking

ChickenParm: Dim1=0.91 · Dim2=0.04 · Dim3=0.07 · Dim4=0.02

SVD: OUTPUT

Input:

M_sparse shape=(10000, 5000) nnz=198,421 density=0.4%

Fit — randomized SVD, $k = 128$:

variance_explained = 91.3%

top singular values: [142.3, 118.7, 97.1, 84.2, ...]

Output:

M_latent shape=(10000, 128) dtype=float32 density=100%

>> execution_success

TF-IDF ✓ → UMAP ✓ → SVD ✓ → K-NN

K-NEAREST NEIGHBORS

K-Nearest Neighbors

Efficiently finding the closest recipes in vector space.

K-NN: FINDING SIMILAR RECIPES

Every recipe is now a 128D coordinate. "Similar recipes" means "nearby coordinates."

A user clicks "Chicken Parmesan."

We need the K recipes whose vectors sit closest to $\tilde{v}_{\text{ChickenParm}}$.

Brute-force: compare the query against every recipe.

$$\text{Cost} = O(N \cdot D) = O(10\{, \}000 \times 128) = 1.28 \times 10^6 \text{ ops/query}$$

Fast enough now — but at 1M recipes $\times$ raw 5K dims: 5×10^9 ops.

We need a smarter structure.

K-NN: FINDING SIMILAR RECIPES

Every recipe is now a 128D coordinate. "Similar recipes" means "nearby coordinates."

The insight: if we spatially index the vectors, we can skip most comparisons.

Recipes that are far away geometrically **can never be close neighbors**

– so we don't check them.

We use a **KD-Tree** – a binary tree that partitions vector space into nested regions.

Build once. Every query becomes $O(\log N)$ instead of $O(N)$.

KD-TREE: SPLITTING SPACE

Recursively cuts the search space in half along the most spread-out dimension.

At each node, split at the median of the widest dimension:

$$\pi_{d,t} = \{x \in \mathbb{R}^D : x_d = t\}, \quad d = \arg \max_j \text{Var}(x_j)$$

Left: $x_d < t$. Right: $x_d \geq t$. Recurse until leaves hold ≤ 40 recipes.

Build cost: $O(DN \log N)$ – paid once.

For 10,000 recipes: 14 branch decisions to reach a leaf.

KD-TREE: SPLITTING SPACE

Recursively cuts the search space in half along the most spread-out dimension.

Pruning – the key trick:

After checking the nearest leaf, walk back up the tree.

At each branch, measure the distance to the splitting plane:

$$d_{\pi} = |\tilde{v}_{\text{query}[d]} - t|$$

If $d_{\pi} \geq$ current best distance: **skip the entire subtree.**

For clustered recipe data: **96% of the tree is pruned per query.**

312 nodes visited out of 10,000.

K-NN: DISTANCE METRIC

We use cosine similarity – it measures the angle between vectors, not their length.

Euclidean – measures gap

$$d = \sqrt{\sum_i (A_i - B_i)^2}$$

A recipe scaled to double portions looks far from its twin despite identical flavor.

Fooled by scale.

Cosine – measures angle

$$\text{sim}(A, B) = \frac{A \cdot B}{\|A\| \cdot \|B\|}$$

Same flavor ratios = 1.0
regardless of recipe size.
Scale-invariant.

L2-normalize all vectors: $\hat{v} = v / \|v\|$.

On the unit sphere, cosine similarity equals Euclidean distance.

One step – best of both metrics.

K-NN: OUTPUT

```
find_neighbors(ChickenParm, K=5, metric=cosine)
```

```
[cos=0.97] Eggplant Parmesan → [13.8, -3.5, 8.4, ...]  
[cos=0.94] Chicken Milanese → [14.9, -4.1, 7.6, ...]  
[cos=0.91] Veal Parmesan → [13.1, -3.2, 9.0, ...]  
[cos=0.88] Baked Ziti → [12.4, -5.0, 8.8, ...]  
[cos=0.85] Chicken Marsala → [15.3, -2.9, 7.1, ...]
```

K-NN: OUTPUT

```
find_neighbors(ChickenParm, K=5, metric=cosine)
```

```
>> query_success
```

```
latency: ~2ms | nodes_visited: 312 / 10,000 | pruned: 96.9%
```

TF-IDF ✓ → UMAP ✓ → SVD ✓ → K-NN ✓

SBERT

Sentence-BERT

A second vectorization strategy – this time, biased toward meaning.

SBERT: A SECOND VECTORIZER

TF-IDF vectorizes by token frequency. SBERT is a completely separate model that vectorizes by meaning.

Both produce vectors that feed into the same UMAP visualization pipeline.

They are two different answers to the same question:

"How do we turn a recipe into a point in space?"

TF-IDF

Counts ingredient tokens.

"Chicken" and "Poultry"

are unrelated axes.

Lexical.

SBERT

Encodes semantic meaning.

"Chicken" and "Poultry"

land near each other.

Semantic.

SBERT: A SECOND VECTORIZER

TF-IDF vectorizes by token frequency. SBERT is a completely separate model that vectorizes by meaning.

Why have both?

They produce different 3D galaxies – different lenses on the same data.

TF-IDF clusters by **shared ingredients**.

SBERT clusters by **culinary meaning and description**.

A recipe called “Braised Short Ribs” and one called “Slow-Cooked Beef in Red Wine”

will be far apart in TF-IDF space – and close together in SBERT space.

Users can switch between views.

SBERT: HOW IT WORKS

SBERT is a fine-tuned BERT model that outputs a single fixed-size vector per sentence.

Standard BERT outputs one vector **per token**, not per sentence. To get a single sentence vector, we average all token outputs — **mean pooling**:

$$u = \frac{1}{n} \sum_{i=1}^n h_i \in \mathbb{R}^{768}$$

Where h_i is the output for the i -th token.
Always 768 dimensions, regardless of input length.

SBERT: HOW IT WORKS

SBERT is a fine-tuned BERT model that outputs a single fixed-size vector per sentence.

How meaning is trained in:

SBERT is trained with triplet loss – similar sentences are pulled together,
dissimilar ones pushed apart:

$$\mathcal{L} = \max(0, \|u_a - u_p\|^2 - \|u_a - u_n\|^2 + \varepsilon)$$

After training: **geometric distance** $\approx$ **semantic distance**.

“Spicy chicken stir-fry” and “fiery poultry wok dish” land close together.

Same concept $\rightarrow$ small angle $\rightarrow$ high cosine similarity.

SBERT: THREE VISUALIZATION MODES

The site runs three separate SBERT models in parallel, each producing a different 3D galaxy.

Ingredients

Mode A

SBERT encodes the **ingredient list** as a sentence.

Clusters by what's in the recipe.

Similar to TF-IDF but meaning-aware —

Recipe Name

Mode B

SBERT encodes only the **recipe title**.

Clusters by dish concept and cuisine.

“Chicken Tikka” and “Chicken Curry”

Combined

Mode C

SBERT encodes **name + ingredients** together.

Balances both signals.

The default view — captures both what

"olive oil" and
"EVOO" land
together.

cluster even if
ingredients differ.

a dish **is** and what
it **contains**.

SBERT: SAME PIPELINE, DIFFERENT LENS

Each SBERT mode feeds directly into the same UMAP → Three.js pipeline as TF-IDF.

All three SBERT models output 768D vectors.
UMAP compresses each set independently into its own 3D coordinates.

Mode A (Ingredients): `v_ChickenParm = [+11.3, -2.1, +6.8]`

Mode B (Name): `v_ChickenParm = [+8.7, +4.2, -1.3]`

Mode C (Combined): `v_ChickenParm = [+12.9, -1.7, +5.2]`

Three different maps. Same recipes. Different groupings depending
on which signal matters most to the user.

SBERT: SAME PIPELINE, DIFFERENT LENS

Each SBERT mode feeds directly into the same UMAP → Three.js pipeline as TF-IDF.

```
>> all_modes_ready
```

TF-IDF vectorizer	→	UMAP	→	3D galaxy (token frequency)
SBERT / Ingredients	→	UMAP	→	3D galaxy (ingredient meaning)
SBERT / Name	→	UMAP	→	3D galaxy (dish concept)
SBERT / Combined	→	UMAP	→	3D galaxy (full context)

The user selects a mode. The visualization updates.
Same K-NN search engine underneath — only the vector space changes.

K-NN EVALUATION

[Returning to the primary text...](#)